

Praveen Kumar Reddy M

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EDUCATION

MSc in Technology Management | Lindsey Wilson University | USA **8/2023 – 5/2025**
Courses: Computer and Network Security, Business Management, Cyber Law, Information Systems, Emerging technologies
BSc in computer science | Babu Jagjivan Ram Degree College | India **8/2019 – 5/2022**
Course: Java, Data Structures and Algorithms, Web Technologies, OOPs, Database Management Systems

SKILLS

Programming: Java, SQL, CSS, HTML, Scala, Groovy, Go, Bash, Shell Scripting.
Frameworks: TensorFlow, PyTorch, scikit-learn, XGBoost, LangChain, Hugging Face Transformers, spaCy.
Technologies: AWS, Azure, GCP, MySQL, PostgreSQL, Oracle, MongoDB, Spark, Kafka, Vertex AI, Bedrock.
Developer Tools: VS Code, PyCharm, IntelliJ IDEA, Git, GitHub, GitLab, Bitbucket, Jira, Terraform.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | CVS Health | Texas, USA **Aug 2024 – Present**

- Python, PyTorch, TensorFlow, scikit-learn, AWS SageMaker, LangChain, Bedrock, MLflow, Airflow, Kubernetes.**
- Designed and deployed ML-driven prescription and claims anomaly detection models, reducing false-positive alerts by 35% using ensemble learning and explainable AI.
- Built retrieval-augmented generation workflows with LangChain and AWS Bedrock, integrating medical text embeddings for context-aware recommendations in pharmacy analytics.
- Automated training, deployment and monitoring pipelines in SageMaker using MLflow for versioning and Airflow DAGs for scheduling, achieving 90% automation coverage and also implemented MLOps pipelines using sagemaker.
- Collaborated with backend teams to integrate ML inference endpoints via Spring Boot microservices, improving prediction latency by 40%. Built CI/CD-driven MLOps workflows integrating GitHub Actions, Docker, and Kubernetes, enabling seamless retraining and model rollout across environments with zero manual intervention.
- Architected and deployed GenAI-powered prescription validation and RAG-based pharmacy recommendation systems using LangChain, Bedrock, OpenAI APIs, and Hugging Face Transformers, reducing manual review time by 50%.

Graduate Research Assistant | Lindsey Wilson College | Kentucky, USA

Jan 2024 – Aug 2024

- AWS Lambda, Step Functions, Airflow, MySQL, Docker, Pandas, Matplotlib, EC2**
- Developed predictive analytics pipelines for academic performance forecasting using TensorFlow regression models, increasing model accuracy to 92% and featured engineering pipelines in Airflow to automate student data ingestion
- Built model performance dashboards using Matplotlib and AWS QuickSight for non-technical stakeholders.
- Created RESTful Flask APIs to serve ML models via AWS and integrated CI/CD for automatic retraining with Step Functions.
- Optimized MySQL query execution and integrated model predictions with institutional systems in real time triggers.

Machine Learning Engineer | Nationwide Mutual Insurance & Banking | Hyderabad, India

Jan 2022 – Aug 2023

- XGBoost, FastAPI, Oracle, PostgreSQL, Docker, Jenkins, Pandas, NumPy, Power BI, Kafka.**
- Designed and implemented risk prediction and fraud detection models using XGBoost and Logistic Regression, achieving a 45% improvement in fraud identification accuracy. Built ETL pipelines with Kafka, FastAPI, PostgreSQL for model data ingestion, automating data cleaning and transformation using Pandas and also applied explainable AI to provide interpretability on credit risk models.
- Deployed trained models as RESTful APIs via FastAPI integrated with the claims processing microservices, enabling real-time risk prediction at scale.

PROJECTS

Enterprise E-Commerce Microservices Platform

- Cloud Bootcamp, Python, Spring Boot, Spring Cloud, REST APIs, AWS, Docker, Kubernetes, Jenkins, Terraform, MySQL.**
- Enhanced the e-commerce platform with an AI-driven recommendation engine using Python and AWS Lambda
- Implemented RESTful APIs with Spring Cloud Gateway for inter-service communication and integrated MySQL
- Integrated AWS Lambda, Step Functions, Airflow, DynamoDB for asynchronous event & implemented API security

Resume Optimizer Web Application

- Personal Project - Java 17, Spring Boot, Spring MVC, REST APIs**
- Developed and deployed a Spring Boot web application with RESTful APIs and MySQL on AWS EC2, Step Functions, Airflow, DynamoDB enabling dynamic resume optimization and improving processing and deployment efficiency by over 60%.

Analyzing and Predicting NYC Taxi Traffic - REST APIs, MySQL, Spring Data JPA, AWS, Docker, Thymeleaf, Chart.js

- Developed a Spring Boot-based analytics dashboard to visualize and monitor real-time traffic patterns across NYC using RESTful APIs and MySQL-backed data services.
- Implemented data ingestion and aggregation modules using Spring Data JPA and scheduled background jobs.